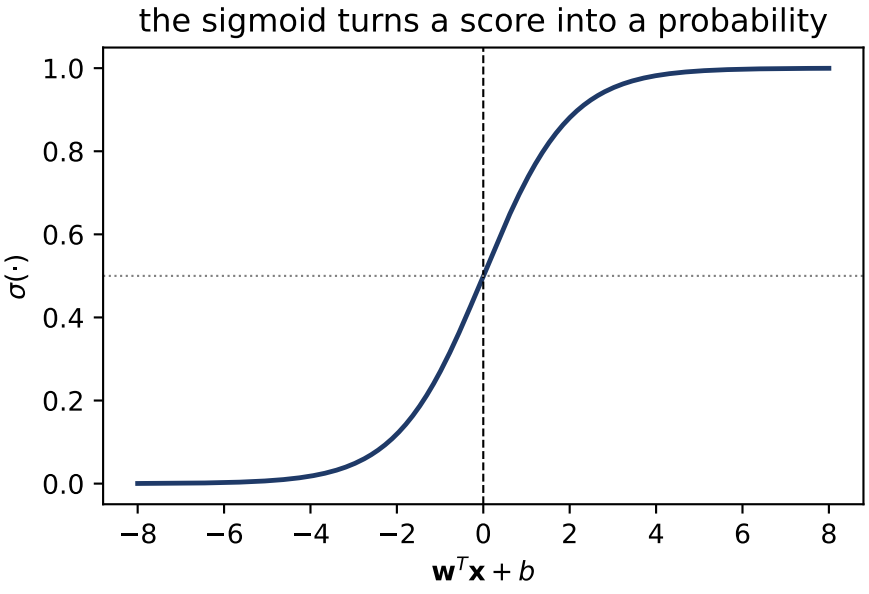
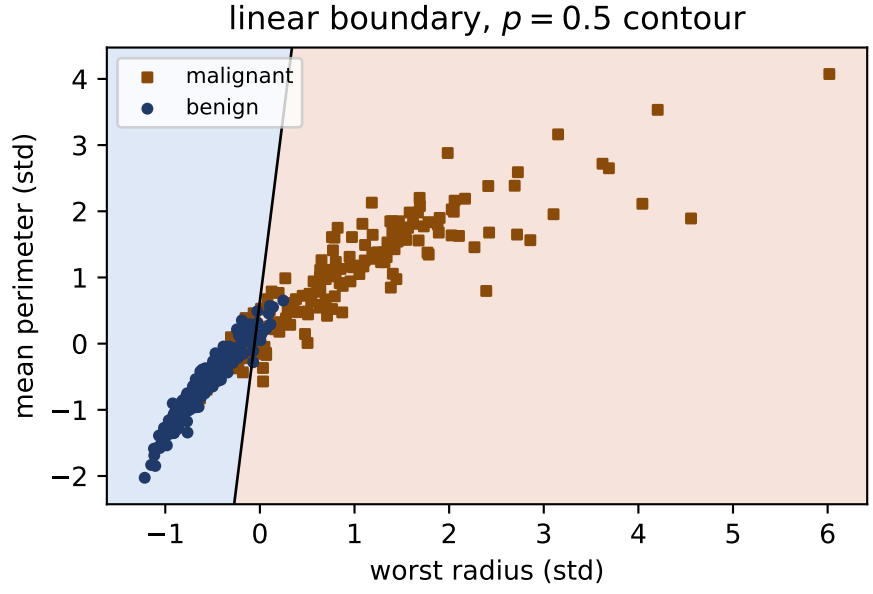

the sigmoid turns a score into a probability
1.0
0.8
0.6
0.4
0.2
0.0
$\sigma(\cdot)$
−8
−6
−4
−2
0
2
4
6
8
$\mathbf{w}^T\mathbf{x} + b$


linear boundary, $p = 0.5$ contour
malignant
benign
mean perimeter (std)
4
3
2
1
0
−1
−2
−1
0
1
2
3
4
5
6
worst radius (std)